

AirBnb Data Analysis

1. Executive Summary

This project analyzed Airbnb listing data from twelve cities worldwide to identify factors associated with guest demand. Demand was measured using review activity during the previous twelve months. The analysis found that lower-priced listings received higher demand, entire homes and apartments were the most popular property type, and listings with minimum stays of two to three nights attracted the highest review activity.

2. Business Problem

Understanding which listing characteristics influence guest demand can help hosts make more informed pricing and listing decisions. This analysis was conducted to identify common attributes associated with higher-performing Airbnb listings across multiple global markets.

3. Business Question

The business questions go as such:

1. Which property types appear to have the strongest demand based on review volume?
2. What listing characteristics are associated with higher guest demand?
3. How does pricing influence occupancy?

4. Data Overview

The data used was twelve different CSVs with Airbnb listing data in each one. Each CSV is from different cities around the world. The final dataset contained 148,725 Airbnb listings collected from twelve global cities. Only a select few columns are used. Columns: room_type, price, minimum_nights, reviews_per_month, number_of_reviews_ltm.

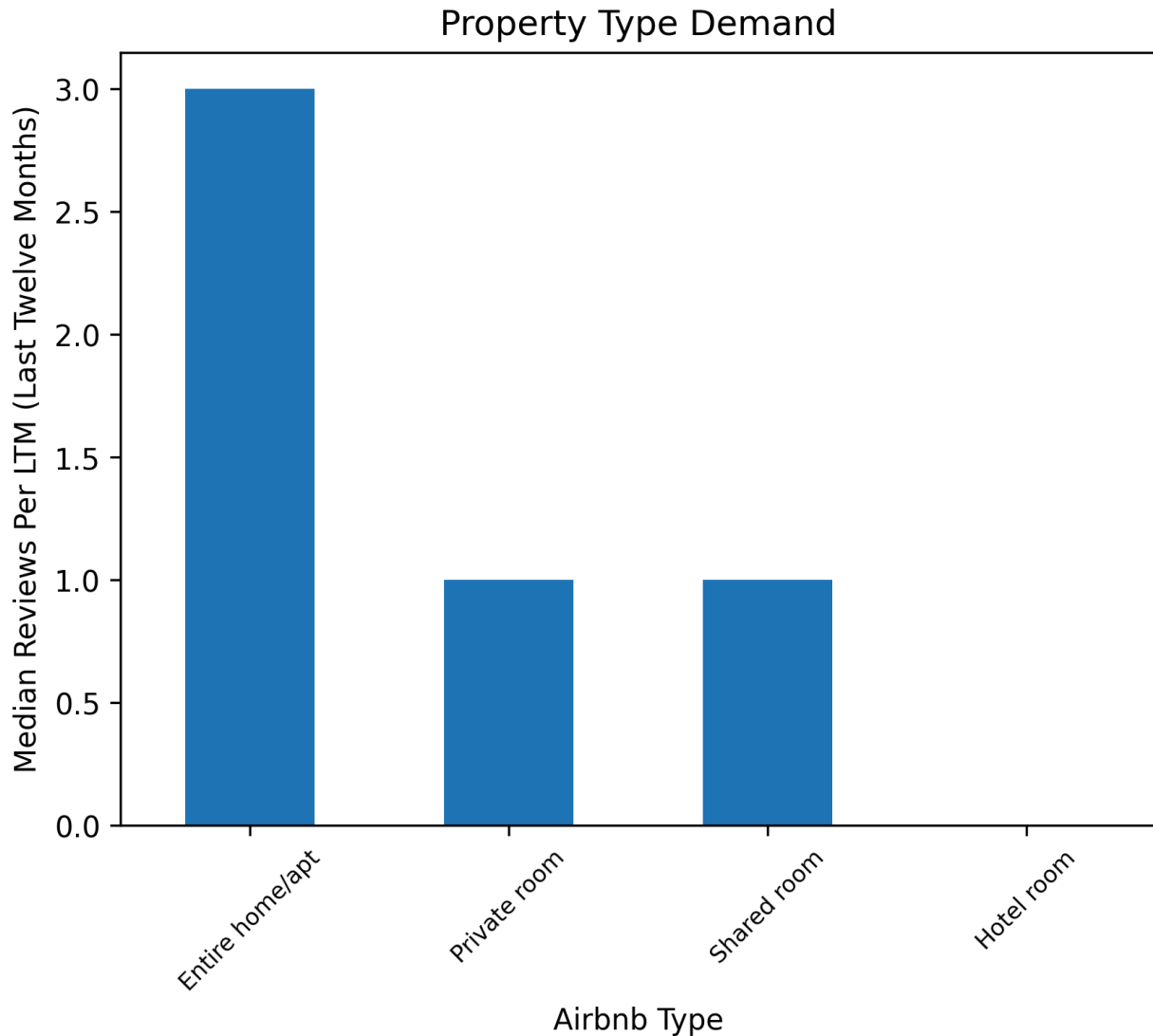
5. Methodology

- Guest demand was measured using number_of_reviews_ltm.
- Median values were used throughout this project due to heavy data skew by outliers found in exploratory data analysis and summary statistics.
- Listings with availability_365 equal to 0 were excluded from analytic operations to focus on actively available listings. .
- The price column was separated into quartile sized buckets using pandas qcut().

6. Analysis & Findings

6.1 Property Type Demand

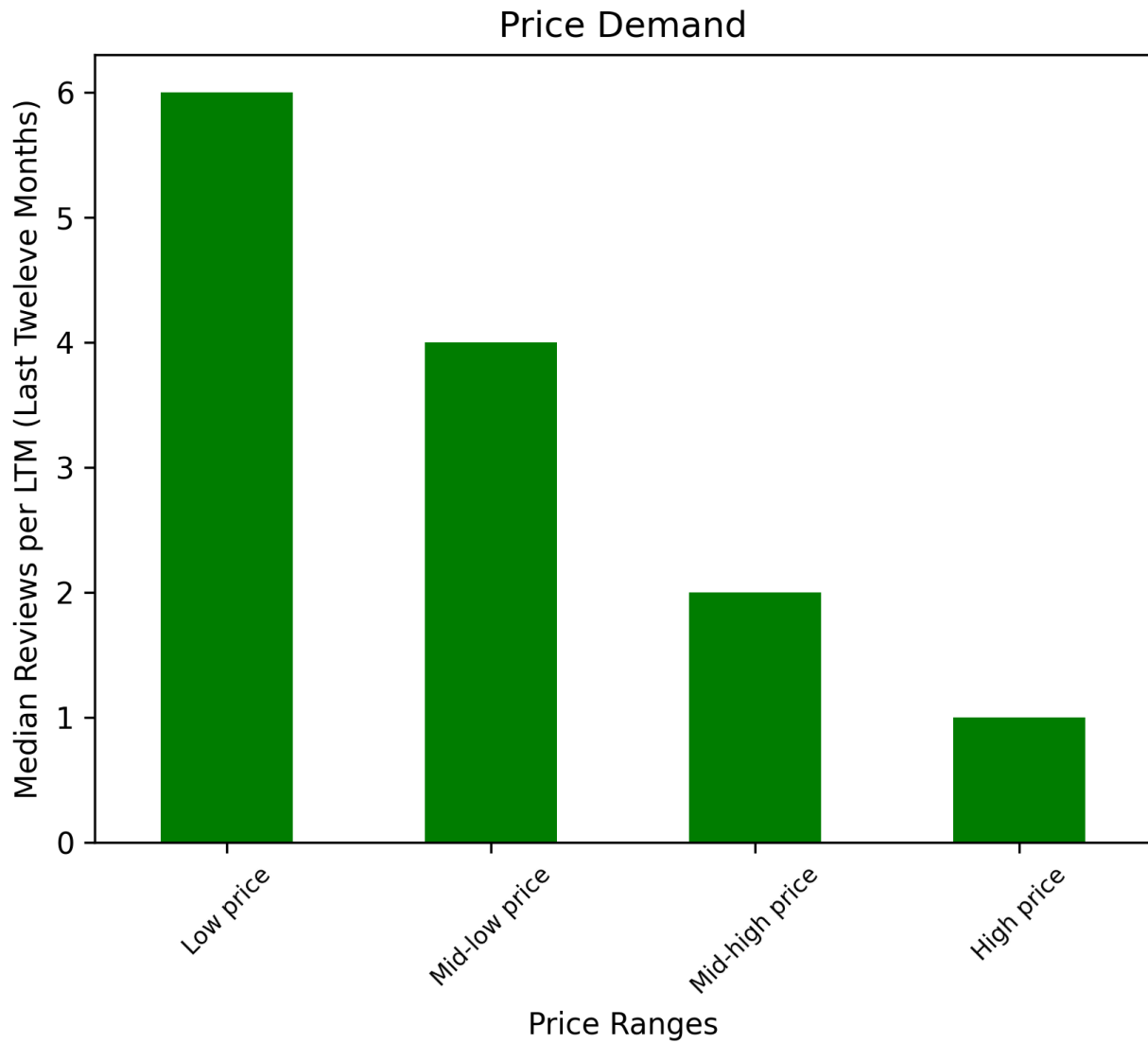
For business question #1, number_of_reviews_ltm and room_type columns were used to figure out which properties which property types appear to have the strongest demand.

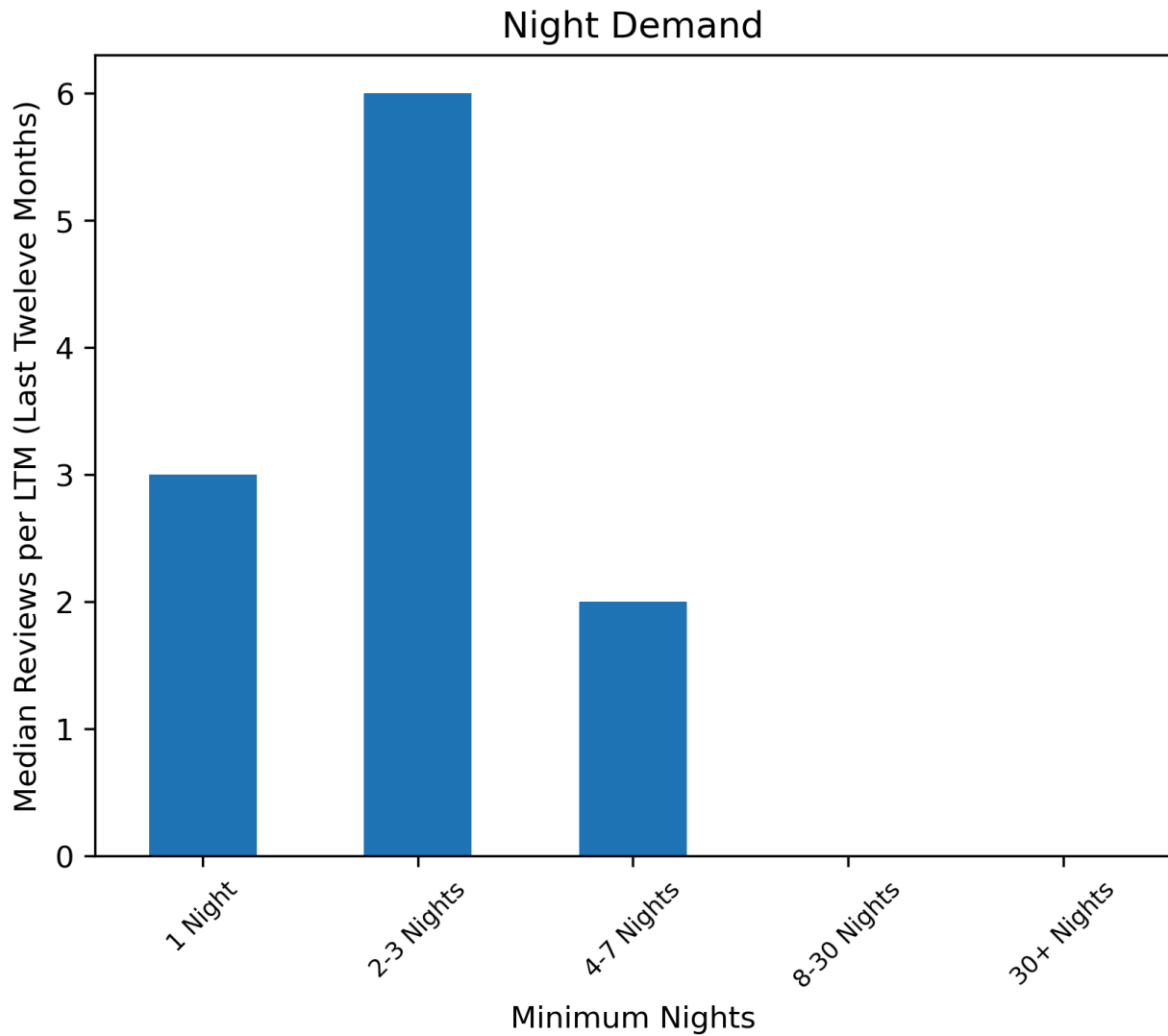


Entire homes and apartments demonstrated the highest median review activity during the previous twelve months, indicating stronger demand relative to other room types.

6.2 Listing Characteristics

For business question #2, listing characteristics has to be determined with proxies. Room_type, price, and minimum_nights were used to highlight what characterized guests' demands. Listing property type demand has already been visualized.

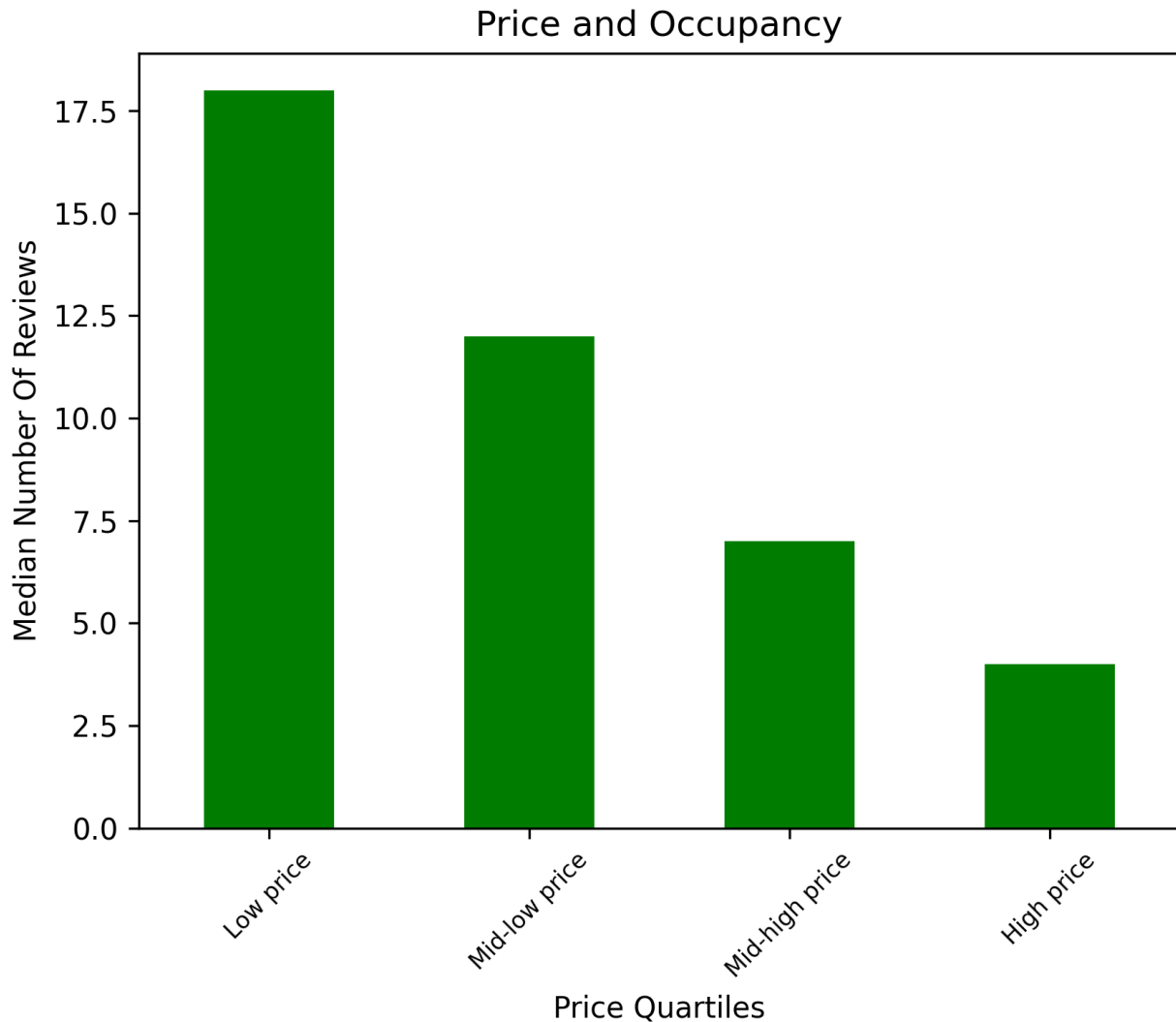




Listings associated with the highest demand were typically entire homes or apartments, were positioned within lower price tiers, and required minimum stays of two to three nights.

6.3 Pricing and Occupancy

For business question #3, determining how pricing influenced occupancy was done through the proxy `number_of_reviews`.



With the chart above we can see that guest demand leans heavily towards low priced listings. Commonalities to the first chart shown are present, confirming price demand.

7. Recommendations

Hosts should evaluate pricing strategies relative to local competition, as lower-priced listings consistently demonstrate higher demand. Flexible booking policies, particularly minimum stay requirements of two to three nights, may improve listing attractiveness.

8. Limitations

In this project we did run into some limitations with our data. For starters, calculating demand by reviews. Demand was estimated using review activity rather than actual booking data. As a result, review counts serve as a proxy for demand rather than a direct measure of occupancy.

Lastly, considering this was a global analysis, there are many regional differences (conversion of prices).

9. Conclusion

This analysis identified several characteristics associated with higher Airbnb demand across twelve global markets. Entire homes and apartments demonstrated the strongest review activity, lower-priced listings consistently attracted greater demand, and shorter minimum-night requirements were associated with improved performance. While review activity was used as a proxy for demand, the findings provide useful insights into listing characteristics that may contribute to Airbnb success.